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EMIRATES INTERNATIONAL SCHOOL, MEADOWS

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QUANTUM ALGORITHM-BASED FEATURE SELECTION FOR MULTI-SENSOR FUSION IN AUTONOMOUS VEHICLE PERCEPTION SYSTEMS

Research Proposal submitted to the International Academic Research Competition (IARCO) 2026

Research Topic: Computer Science

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DUBAI, UNITED ARAB EMIRATES
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1 INTRODUCTION

The modern design of autonomous vehicles (AVs) implies the utilization of various multi-sensors setups for safe navigation in the dynamic environment [2, 3]. Thanks to simultaneous acquisition of LiDAR and camera data, an AV can get a complete 3D visualization of its surroundings, including the road, other vehicles, and pedestrians [1, 2]. At the same time, real-time processing of large amounts of data acquired from various sensors creates an extra computational load, leading to increased latency and posing a threat to the vehicle's capability to make quick decisions about the emergency braking or lane change [2, 3].

One of the main causes of this computational delay is represented by high-dimensional sensor data [2, 4]. Since new sensors are added by the engineers to compensate for blind spots, the resulting dataset is overpopulated by overlapping and redundant features [2, 4]. Forcing machine learning algorithms to work with repetitive spatial information (such as the repeated acquisition of object's length, width, and height) results in an excessive usage of computational power and batteries without improving the classification [4, 11]. Searching for the optimal balance between perception accuracy and high processing speed is the most urgent problem of autonomous driving at the moment [2, 3, 10].

This challenge can be addressed through the implementation of the feature selection strategy allowing to choose only the most significant spatial attributes and feed them to machine learning models [4, 11]. The problem of the classical feature selection strategy is based on such techniques as Mutual Information ranking and Recursive Feature Elimination [4, 5, 11]. It should be noted that classical strategies are usually oriented towards the evaluation of the features separately or in accordance with a pre-defined path [4, 11]. Because of this approach, the classical methods are not capable of providing an overview of the whole dataset, resulting in the inability to exclude the redundant data and select the optimal set of features [4, 6]. The introduction of quantum computing allows for solving this problem in a brand-new way [7, 8]. Framing the feature selection as a Quadratic Unconstrained Binary Optimization (QUBO) problem and resolving it with the help of Quantum Approximate Optimization Algorithm (QAOA), it becomes possible to assess the relevance of the features and penalize the redundancy simultaneously [5, 6, 7].

While this approach looks promising theoretically, the evaluation of QAOA-based feature selection for autonomous vehicles in the framework of Noisy Intermediate-Scale Quantum (NISQ) restrictions is still a relatively unexplored area [3, 8, 10]. It should be admitted that there is no evidence proving the possibility of reducing the high-dimensional feature sets to compact size by means of QAOA without deteriorating the accuracy of classification or increasing the processing time [3, 6, 8]. This research project is aimed at the analysis of applicability of QAOA-based QUBO optimization for spatial feature selection based on multi-sensor LiDAR and camera data from the nuScenes [1] dataset. Comparing the performance of secondary perception models (such as Support Vector Machines, Random Forests, and Logistic Regression) when trained on QAOA-selected features with filter and wrapper baseline methods will show whether or not quantum feature compression can be used in autonomous vehicles [1, 4, 11].

Problem Statement: The current classical feature selection methods utilized in autonomous vehicle perception pipelines do not take into account the global interactions of features and are not able to eliminate spatial redundancies properly, generating high-dimensional feature sets and

decreasing the speed of real-time processing and wasting computing resources without increasing object classification accuracy [2, 4, 6, 11].

The need for addressing such a perceptual bottleneck becomes increasingly urgent considering the advent of Level 4 and Level 5 autonomous cars and the associated problems of excessive amounts of sensor data, which create processing delays in embedded processors. This work will help car developers through creating a method that can be used to efficiently reduce the dimensionality of spatial data while maintaining computational efficiency and reducing energy costs without compromising safety. Moreover, it gives the autonomous perception field an opportunity to test NISQ-based QAOA methods.

2 RESEARCH QUESTION

Central Research Question: How effectively can quantum algorithm-based feature selection, specifically Quantum Approximate Optimization Algorithm (QAOA) applied to Quadratic Unconstrained Binary Optimization (QUBO) formulations, optimize the trade-off between feature subset dimensionality and object perception performance in autonomous vehicle sensor fusion pipelines compared to classical selection methods, such as Principal Component Analysis (PCA)?

We hypothesize that if multi-sensor feature selection is formulated as a Quadratic Unconstrained Binary Optimization (QUBO) problem solved via the Quantum Approximate Optimization Algorithm (QAOA), then it will give a more compact feature subset (k=3) that retains classification performance that's comparable to or exceeding classical filter/wrapper methods (k=5), because QAOA evaluates global inter-feature correlations and pairwise redundancy penalties simultaneously within the energy Hamiltonian, eliminating collinear spatial features that classical greedy methods retain.

3 LITERATURE REVIEW

Perception systems in autonomous vehicles (AVs) today use several sources of input data, such as 3D LiDAR point clouds and videos captured by several cameras to build the map of the dynamic environment [1, 2]. However, continual processing of the large number of high-resolution spatial data causes computational load [1, 2]. This causes delay and brings considerable risks to the safety of the autonomous vehicle such as braking and avoiding the collisions [2, 3].

In order to reduce the computational overhead on the board, the features are reduced before passing to classifiers [4, 11]. Classical dimensionality reduction techniques (PCA, t-SNE, UMAP) reduce the variance of the input data in the low-dimensional spaces [4]. However, they transform physical variables (such as size of the bounding box) into abstract latent features that do not have any physical meaning and cause additional computation cost for every inference process [1, 4].

Feature selection technique helps to preserve the meaning of the features by selecting the subset of k features [4, 11]. Classical filter approaches such as Mutual Information Ranking evaluate the individual feature relevance (α_i) separately from one another and, thus, do not consider the

inter-feature relationships (β_{ij}) between the features and pick up a lot of redundant features creating the cluster in the spatial space [4, 5]. Wrapper approaches, such as SVM-RFE, evaluate the subsets of features in an iterative way but poorly scale and are stuck in the local optima [4, 11]. In turn, the feature selection task in quantum computing can be represented as the QUBO energy matrix problem and solved with Quantum Approximate Optimization Algorithm (QAOA), which evaluates also global feature relevance (α_i) and pairwise redundancy penalties (β_{ij}) simultaneously to isolate the target k-feature subset [5, 6, 7].

Research Gap

Despite having significant theoretical potential, it lacks practical proof of whether QAOA-based feature selection has the ability to downscale the large high-dimensional spatial feature space to an extremely small feature subset (k=3) without losing object classification accuracy or inference time in the NISQ environment [3, 8, 9, 10].

4 METHODOLOGY

4.1 Research Design

The experiment described below utilizes the experimental and quantitative paradigm for the evaluation of quantum-assisted spatial feature selection. As illustrated in Figure 1, the pipeline of the proposed experiment consists of five main stages, that are namely: (1) Data Ingestion & Preprocessing, the extraction of spatial features and their pre-processing result in the construction of a clean feature matrix X [1, 4] out of nuScenes bounding boxes; (2) QUBO Matrix Formulation, non-parametric Mutual Information is used to derive the cost function Hamiltonian (H_C) for feature selection that takes into account both the relevance and spatial redundancy of individual features [5, 6]; (3) QAOA & Baseline Feature Selection, the QUBO problem is solved by simulating the QAOA quantum circuit and results in an ultra-compact feature subset (k=3). In addition, classical baselines (Mutual Information filter and SVM-RFE to name) determine feature subsets (k=3 and k=5) [4, 7, 11]; (4) Downstream Classifier Training, second level perception classifiers (including Support Vector Machine, Random Forest, and Logistic Regression) are trained with the help of the selected feature subsets [11]; and (5) Benchmark & Latency Evaluation, classification accuracy and F1 score and latency are calculated using 5-folds cross validation split [2, 3, 4].

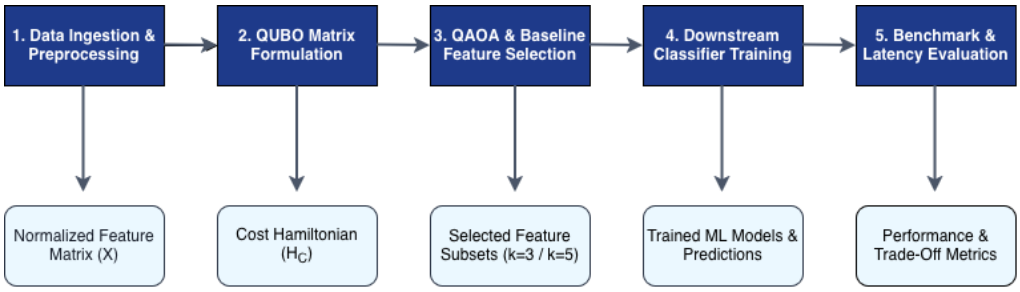


Figure 1. *Proposed methodology pipeline detailing process stages and corresponding output artifacts.*
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4.2 Data Collection

The first source dataset contains 10 features that were chosen among dimensions, positions and velocities of the 3D bounding boxes in LiDAR and camera data streams: 3D bounding box dimensions (dx, dy, dz), spatial coordinates (x, y, z), velocity estimates (vx, vy), LiDAR points number (N_{pts}), and camera aspect ratio (AR) [1, 2].

Stratified sampling of M = 10,000 labeled bounding boxes is carried out for key object categories (Vehicles, Pedestrians, Cyclists, Barriers) [1]. The features are normalized to the ranges [0, 1] with the help of Min-Max scaling [4]. The QUBO cost Hamiltonian is formulated by combining non-parametric estimation of Mutual Information [5], binary optimization mapping [6], and the QAOA framework [7] as follows:

$$H_C(x) = -\sum_{i=1}^{N} \alpha_i x_i + \sum_{i=1}^{N}\sum_{j>i}^{N} \beta_{ij} x_i x_j + \lambda\left(\sum_{i=1}^{N} x_i - k\right)^2 \qquad [5, 6, 7]$$

Where $\alpha_i = I(X_i; Y)$ and $\beta_{ij} = I(X_i; X_j)$ are measures of feature relevance and pairwise redundancy using k-nearest neighbors Mutual Information estimator [5]. The parameter λ ensures that the cardinality constraint of the subset (k=3) is satisfied in the QUBO formulation [6]. The cost operator $H_C(x)$ is optimized using QAOA [7] through Qiskit simulators for p=2 and COBYLA [7, 8].

4.3 Data Analysis

All preprocessing procedures, data handling and statistical tests are performed with the use of Python 3.10 along with scikit-learn (v1.3.2), qiskit (v0.45.2), numpy (v1.26.4), and pandas (v2.1.4) packages. Evaluation criteria such as Accuracy, Macro F1-Score and Log Loss for classification performance are calculated with 5-fold stratified cross-validation [4, 11]. Finally, in order to prove the hypothesis about the non-inferiority of the subset of QAOA (k=3) compared to the original one, the paired two-tailed t-tests using a statistical significance threshold of with $\alpha_{stat} = 0.05$ (are used for comparing cross-validation accuracies [4, 11]. The differences in execution time (milliseconds) are evaluated by the means of non-parametric Wilcoxon signed-rank tests [2, 3].

5 EXPECTED RESULTS

The key outcome of this research would be experimental demonstration that the Quantum Approximate Optimization Algorithm (QAOA) is capable of compressing high-dimensional spatial feature spaces into ultra-compact representation (k=3). It is expected that the subsequent classification models (SVM, Random Forest, Logistic Regression) trained on the QAOA-selected k=3 subset would demonstrate prediction accuracy and F1-scores comparable to or better than those of classical k=5 baseline approaches (Mutual Information Filter and SVM-RFE) while significantly reducing the inference latency.

Innovation, Urgency and Impact

- **A novel methodological approach:** Uses QUBO formulation to find spatial features using parameterized quantum circuits, which allows the determination of the importance of the global feature (α_i)and spatial redundancy (β_{ij}) simultaneously in an energy matrix without any classical computation steps.
- **Importance of research at this time:** Level 4-5 of autonomous vehicles generate massive point-cloud and image data streams, creating chip processing lags that threaten real-time avoidance of collisions. Modern quantum simulation tools (like Qiskit) and open-access datasets (like nuScenes) now make quantum-classical optimization viable for autonomous perception.
- **Impact of research & beneficiaries:** Allows autonomous systems designers to minimize latency, power consumption, and thermal throttling of edge computing devices. Offers the perception field with a benchmark for using hybrid quantum algorithms in the perception pipeline of vehicles.

6 POTENTIAL LIMITATIONS

Despite all the rigor of the proposed research that has been explained, the following limitations may apply to it in terms of practicality of application. There are problems of restricting the number of features N = 10 and the circuit depth p = 2 when applying QAOA to classical quantum simulators due to speed optimization requirements, which have been solved by employing Qiskit statevector simulators and layer-wise parameter initialization to prevent barren plateaus. The second problem is that there is no use of temporal information when performing feature extraction, which is done based on spatial data from nuScenes without extreme weather, but this problem is balanced by stratification when choosing 10,000 bounding boxes. The final limitation is the use of classical ML classifiers in the secondary evaluation and not deep neural networks.

7 CONCLUSION

The contemporary issue facing autonomous vehicle perception systems concerns the trade-off between accurate perception and processing time. With high-density sensors gathering high-dimensional spatial data, classical feature selection algorithms have proven inefficient at discarding spatial redundancy without introducing any processing delay. In this work, we intend to tackle this issue by assessing the possibility of formulating multi-sensor feature selection in a QUBO problem and solve it using QAOA in order to identify an ultra-compact set of features (k=3) that preserve the performance of subsequent classifiers yet outperform classical methods (k=5). Using 3D spatial data from nuScenes, we construct non-parametric Mutual Information matrices, simulate QAOA circuits, and evaluate downstream classifiers using 5-fold cross-validation.

The expected contribution of this work is empirical evidence that quantum-assisted optimization can collapse collinear sensor features. In conclusion, this approach represents an important step toward solving the problem of processing delay in autonomous driving perception pipelines.

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